

ORIGAMI User Manual

Stereo triple wave folder · Jidai Collection · Martial Systems

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1. Overview

ORIGAMI is a stereo wave folder. A wave folder takes the parts of a waveform that would rise past a limit and folds them back down, so a plain wave turns into one rich in harmonics. ORIGAMI has three folding stages in series under one WAVE knob. Turn WAVE up and stage 1 comes in first, then stage 2, then stage 3. The sound goes from a warm edge to dense, glassy harmonics.

Around the folder are the tools that make it easy to use on real material:

- **GAIN** and a **pre-fold VCA**, so you can set how hard the signal hits the folder, by hand, from an envelope follower or from a control voltage.
- **EMPH** (emphasis), which boosts the highs before the folder and removes exactly that boost after it.
- **SYM** controls that tilt the fold for even harmonics.
- **VC 1, VC 2 and VC 3**, one modulation input per stage, fast enough for audio-rate modulation.
- **LEVEL COMP**, which keeps the output level steady while you sweep WAVE.
- **MIX**, a dry/wet blend in which the dry signal is delayed to line up with the wet one.

At WAVE 0 with the other controls at their defaults, ORIGAMI passes your audio through unchanged, bit for bit.

ORIGAMI runs as a VST3 effect, as a standalone app, and as a device in the JIDAI RACK. In the rack, its jacks patch into the other Jidai devices. Left and right share every control, and the level detectors listen to both channels together, so the stereo image holds.



2. Quick start

1. Install the plugin (see section 7) and insert ORIGAMI on an audio track or bus.
2. Play some audio through it. With **WAVE** at 0 you hear the track unchanged.
3. Raise **WAVE** slowly. Below about a third of the way up you are in stage 1, a warm fold. Higher up, stages 2 and 3 join in.
4. Turn **SYM** away from the centre for a lopsided fold with more even harmonics.
5. Turn **MIX** down to blend the folded sound with the dry signal.
6. If the sound is too tame or too wild, adjust **GAIN** on the INPUT side. It sets how hard the signal hits the folder.

To start from a sound designed for your source, click the preset name at the top of the window and pick one from the bank for that source (section 6).

3. Panel reference

3.1 Using the controls

Action	Result
Drag a knob up or down	Changes its value. Hold Shift for fine control (5× finer).
Mouse wheel over a knob	Small steps. Hold Shift for even smaller steps.
Double-click a knob	Returns it to its default.
Click a switch or selector	Steps to the next position.
Right-click a selector	Steps back one position.
Click a page tab	Shows that page.

Every control is a plugin parameter, so your DAW can automate it.

3.2 Header

Element	What it does
MAIN / STAGES / DYNAMICS / SETUP	Page tabs.
Preset box	Shows the current factory preset as <i>BANK · Name</i> . The arrows step through all presets in order. Click the name for a menu grouped by bank. In the rack, the preset box sits in the jack row, between VC 1 and VC 2.

Element	What it does
Status (top right)	Shows whether 2× quality is on and the current latency in samples, for example 2× OFF · LAT 0.

The jack row along the bottom appears on every page (see 3.7).

3.3 MAIN page



INPUT box

Control	Range	Default	What it does
GAIN	−24 to +12 dB	0 dB	Input level into the folder. More gain means a deeper fold at any WAVE setting.
VCA DEPTH	0–100 %	0 %	How much the pre-fold VCA moves the drive. At 0 % the VCA does nothing. At 100 % the drive follows the VCA source completely.
FOLLOW / CV	switch	FOLLOW	The VCA source. FOLLOW uses the envelope follower on the input. CV uses the sidechain follower (when something feeds the sidechain), or in the rack a cable into the VCA CV jack, which wins over the sidechain. The same switch also appears on the DYNAMICS page.
Meter			Input level, with the follower level underneath.

The drive into the folder is $1 - \text{DEPTH} + \text{DEPTH} \times E$, where E is the follower or the CV (0 to 1). With VCA LAW set to EXP, E is squared first.

WAVE box

Control	Range	Default	What it does
WAVE	0–100 %	0 %	The main fold amount. It sweeps stage 1, then stage 2, then stage 3. At 0, with the trims and EMPH also at 0, the folder is bypassed exactly, wherever SYM is set.
EMPH	0 to +12 dB	0 dB	Emphasis: a high shelf (centred at 1 kHz) before the folder and its exact inverse after it. The fold bites harder on bright material while the overall tone stays balanced.
SYM	−1 to +1	0	Symmetry for all three stages. Off centre, the fold is lopsided and adds even harmonics.
STAGE 1, 2, 3	−1 to +1	0	Trims that push each stage above or below what WAVE gives it.

OUTPUT box

3.5 DYNAMICS page



FOLLOWER box

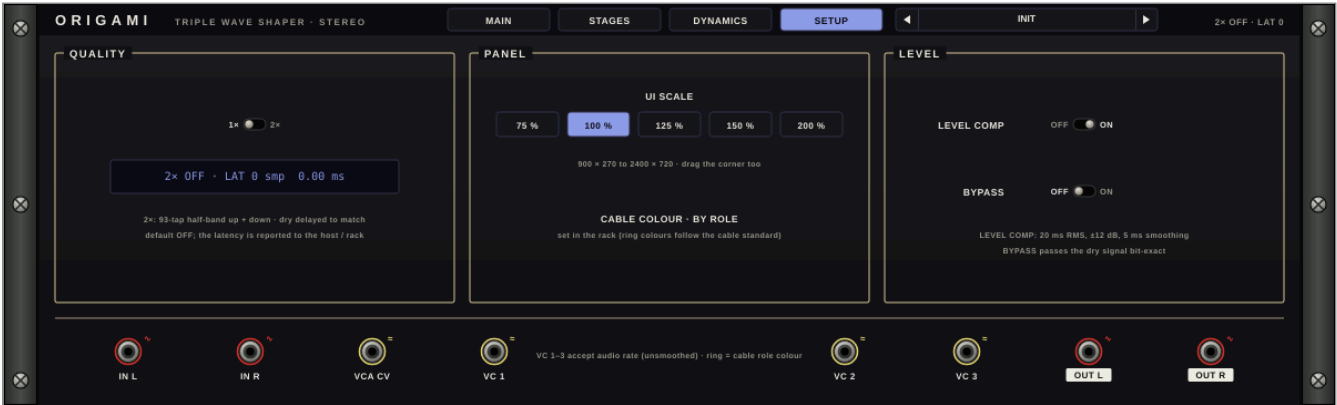
Control	Range	Default	What it does
ATTACK	0.1-50 ms	1 ms	How fast the follower rises.
RELEASE	5-1000 ms	80 ms	How fast the follower falls.
Meter			The follower level. 1.0 corresponds to a full-scale input.

The follower listens to both channels at once and also sets the timing of the plugin's sidechain follower.

VCA box

Control	Range	Default	What it does
CV SCALE	0-2×	1×	Scales the VCA CV. At 1×, 5 V at VCA CV opens the VCA fully.
VCA LAW	LIN / EXP	LIN	EXP squares the VCA source for a snappier response.
VCA SOURCE	FOLLOW / CV	FOLLOW	The same switch as on the MAIN page.

3.6 SETUP page



Box	Control	What it does
QUALITY	1× / 2×	1× (default) has no latency and uses anti-aliased folding; the anti-aliasing softens the very top end a little (about 2 dB at 10 kHz for each stage that is folding, at 48 kHz). 2× runs the folder at twice the sample rate for even less aliasing and adds 46 samples of latency, which ORIGAMI reports to your DAW. The display shows the latency in samples and milliseconds.

Box	Control	What it does
PANEL	UI SCALE 75 %, 100 %, 125 %, 150 %, 200 %	Window size. You can also drag the window's corner. In the rack the size follows the rack.
PANEL	Cable colour	Information only. In the rack, cable colours follow each jack's signal type.
LEVEL	LEVEL COMP off/on	The same switch as on the MAIN page.
LEVEL	BYPASS off/on	Passes the dry signal through bit for bit. At 2× the dry signal is still delayed by 46 samples, so the timing doesn't jump.

3.7 Jack row

Jack	Signal	What it does
IN L, IN R	audio	Audio input. If only IN L is patched, IN R follows it.
VCA CV	CV, 0-5 V	Sets the pre-fold VCA when VCA SOURCE is CV.
VC 1, VC 2, VC 3	CV, audio rate OK	Each stage's modulation input. A cable here wins over the stage's VC source.
OUT L, OUT R	audio	Audio output.

The ring around each jack shows its signal type: red for audio, yellow for CV. Output jacks have a light label. In the plugin the jacks show what they carry. Hover over one to read it. Cables are patched in the JIDAI RACK (see the appendix).

4. Patching

In the plugin

ORIGAMI is a stereo insert effect with one optional extra input:

- **Main input and output:** stereo. A mono input is copied to both sides.
- **Sidechain:** an optional stereo input, off by default. Turn it on in your DAW's routing for the plugin and send another track to it. ORIGAMI then uses the sidechain in two ways:
 - Any stage whose VC source is **SIDECHAIN** is modulated by the sidechain signal.
 - With **VCA SOURCE** on **CV**, a follower on the sidechain (timed by **ATTACK** and **RELEASE**) drives the pre-fold VCA. The louder the sidechain, the harder the input is driven into the folder.

Ideas:

- **Drum-triggered fold:** send a kick to the sidechain, set VC 1 source to **SIDECHAIN**, and raise VC 1 > **AMT**. Each kick pushes the first stage deeper.
- **Self-modulation:** set a VC source to **INPUT**. The signal modulates its own fold at audio rate.
- **Dynamic fold:** raise VCA **DEPTH** with VCA **SOURCE** on **FOLLOW**. Loud notes fold deeper than quiet ones.

In the JIDAI RACK

In the rack every jack in the jack row is live, plus six back-only jacks (see the appendix). Patch audio into IN L/R, CV or audio into VC 1-3 and VCA CV, and take OUT L/R wherever you like. The back-only **SIDECHAIN** > **SC L/R** jacks are the sidechain input. If you patch both, they are averaged to mono.

They work as described above. Signals are exchanged with the other devices sample by sample, so audio-rate modulation from another device works.

5. MIDI and host sync

ORIGAMI doesn't use MIDI notes or the host tempo. It responds the same way at any tempo.

- **Automation:** every control is a host parameter.
 - **Programs:** the factory presets appear in your DAW's program list as *BANK: Name* (for example *RONIN: Acid Grit*). Choosing a program loads that preset.
 - **Latency:** at QUALITY 2×, ORIGAMI reports 46 samples to the host, so your DAW's delay compensation keeps it in time with other tracks.
 - **Saved state:** your DAW project stores every setting. The plugin and the rack device use the same state format, so settings move between them.
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6. Factory presets by bank

There are 33 presets in five banks. Click the preset name in the header for a menu grouped by bank, or step through all of them with the arrows. Every preset has LEVEL COMP on and was checked to stay below full scale on test material. Presets are built into the plugin and the JIDAI RACK, so nothing needs installing. In the rack, the same presets are in the preset box in ORIGAMI's jack row.

The banks are voiced for the other Jidai instruments, but any preset works on any source.

INIT

Preset	Description
INIT	Every control at its default. True bypass. Start here to build your own.

RONIN: basses and leads from a synth voice

Preset	Description
Acid Grit	Medium fold with emphasis and a little symmetry, at 2× quality. A gritty edge for resonant bass lines.
Acid Squelch Fold	Light fold that the envelope follower pushes deeper on every note (VC 1 from FOLLOW).
Reese Growl	Strong lopsided fold with stage 2 nudged up, output lowered. For detuned, growling basses.
Wavefolded Bass	Medium fold with stages 2 and 3 held back, 80 % wet. Fat but controlled.
Lead Bite	Medium fold with strong emphasis, so the fold bites on bright leads.
Pluck Edge	Light fold with the VCA and stage 1 following the envelope. The fold opens on each attack.
Hard Sync-Style Lead	Deep fold with heavy emphasis at 2× quality, for tearing lead tones.
Sub-Safe Bass Drive	Light fold with stage 3 pulled back, 60 % wet. Adds drive and keeps the low end clean.

SHOGUN: drums and drum buses

Preset	Description
Drum Smash	Deep fold with extra gain and emphasis, 70 % wet, output lowered. Crushed drums.

Preset	Description
Kick Fold	Stage 1 follows a fast envelope, so each kick folds as it hits. 60 % wet.
Hat Sizzle	Full emphasis and a medium fold at 2× quality, 50 % wet. Adds sizzle to hats and cymbals.
Drum-Bus Glue	Gentle fold, 80 % wet. Thickens a whole drum bus.
Snare Crack	Emphasis plus stage 2 following a very fast envelope. Adds crack to snares.
Transient Fold	The VCA follows a fast envelope, so transients fold harder than the tails. 50 % wet.
Tom Bloom	Asymmetric fold, with stage 1 following a slow-release envelope. Toms bloom as they ring.
Parallel Drum Grit	Medium fold mixed in at 35 % for parallel grit.

BUSHIDO: folds sequenced from a step sequencer

In this bank VC 1–3 are set to JACK, with depths chosen for a 0–5 V sequencer row. Unpatched, each preset is a moderate fold. In the rack, patch a sequencer's CV outputs into VC 1–3.

Preset	Description
Stepped Fold Sequence	Light fold. VC 1, 2 and 3 each move one stage's amount.
Stepped Symmetry	Medium fold. VC 1 and VC 2 move the symmetry of stages 1 and 2 in opposite directions.
Row C Fold Accent	Light fold. VC 3 makes a strong change to stage 3, good for accents from one row.
Pitch-Tracked Fold	Light fold. VC 1 and VC 2 make small changes, so a pitch CV makes higher notes fold a little more.
Fold Arpeggio	Very light fold with emphasis. VC 1 moves stage 1 a lot, and VC 2 moves stage 2's symmetry.
Gate-Opened Fold	Medium fold with VCA SOURCE on CV. A gate or envelope on VCA CV drives the folder harder while it is high.
Stepped Bite	Medium fold with emphasis. VC 1 moves stage 1, and VC 3 moves stage 3's symmetry.

GENERIC: any source, buses and mixes

Preset	Description
Warm Bus Glue	Very gentle, slightly asymmetric fold. Subtle warmth on buses.
Analog Edge	Light asymmetric fold with a little emphasis, 60 % wet.
Even-Order Bloom	Strongly asymmetric fold, 75 % wet. Rich even harmonics.
Lopsided Fold	Strongly asymmetric fold with extra symmetry on stage 1, output lowered.
Self-Modulated Fold	Stage 1 is modulated by the input itself at audio rate.
Stereo Shimmer Fold	Stages 2 and 3 are modulated by each channel's own signal, at 2× quality. A wider image.
Sidechain Pump Fold	Stages 1 and 2 follow the sidechain. Feed a kick or another rhythm to the sidechain. Without one, a plain light fold.
Full Fold	WAVE fully up with all three stages pushed, at 2× quality, output lowered.
Shattered	Extreme: full fold, maximum gain and emphasis, heavy asymmetry and self-modulation, output well down.

7. DAW setup

ORIGAMI is a VST3 audio effect. You can also run it as a standalone app.

Installing

Copy `ORIGAMI.vst3` into your system's VST3 folder:

System	VST3 folder
macOS	<code>~/Library/Audio/Plug-Ins/VST3/</code> (just you) or <code>/Library/Audio/Plug-Ins/VST3/</code> (all users)
Windows	<code>C:\Program Files\Common Files\VST3\</code>
Linux	<code>~/.vst3/</code>

Then have your DAW rescan its plugins. ORIGAMI is listed under **Martial Systems**.

Using it in any DAW

1. Insert ORIGAMI as an effect on an audio track, instrument track or bus.
2. To use the sidechain, enable the plugin's sidechain input in your DAW's routing and send the trigger track to it.
3. Automate any control from your DAW's automation lanes.
4. Leave the DAW's plugin delay compensation on if you use QUALITY 2×.

Example: FL Studio

1. Close FL Studio, copy the VST3 into your VST3 folder, and open FL Studio.
2. Open **Options > Manage plugins** and click **Find installed plugins**. Mark ORIGAMI as a favourite if you like.
3. In the **Mixer**, select the insert track you want, click an empty effect slot and choose ORIGAMI.
4. For a sidechain, route the trigger track to this insert with a sidechain send, then pick it as the plugin's sidechain input in the plugin wrapper's settings.
5. ORIGAMI reports its 2× latency to FL Studio, whose plugin delay compensation uses it.

Standalone

The standalone app opens your default audio input and output. Choose devices in its audio settings.

8. Specs

Type	Stereo triple wave folder (three stages in series)
Formats	VST3 effect and standalone app
Channels	Stereo in/out (mono input is copied to both sides), plus an optional stereo sidechain input
Stereo	Left and right share every control. Follower and LEVEL COMP listen to both channels.
Signal chain	GAIN > pre-fold VCA > emphasis > stage 1 > 2 > 3 > de-emphasis > DC block (8 Hz) > LEVEL COMP > MIX (dry aligned) > LEVEL
Quality	1×: anti-aliased folding, 0 samples latency. 2×: oversampled, 46 samples latency, reported to the host.
True bypass	WAVE 0 with the other controls at their defaults, or BYPASS on: output is bit-identical to input
Parameters	30, all automatable

Factory presets	33 in 5 banks
Pages	MAIN, STAGES, DYNAMICS, SETUP
UI	75 % to 200 % scale, freely resizable
Rack device	3 U open, 1 U closed. 8 front jacks plus 6 back-only jacks. 0 or 46 samples latency, compensated by the rack.
Signal levels (rack)	Audio ± 5 V (full scale). VCA CV 0–5 V. VC inputs accept audio-rate signals.
Consistency	Built so that every supported system computes the same samples from the same settings

9. Troubleshooting

Problem	What to check
No change in the sound	WAVE is at 0 and the other controls are at their defaults, which is true bypass. Raise WAVE. Also check that BYPASS (SETUP page) is off.
Too loud or clipping (OVER lit)	Lower LEVEL or GAIN. Turn LEVEL COMP on. Extreme presets (Shattered, Full Fold) already lower the output. Lower it more for loud sources.
The fold gets quieter as I turn WAVE up	That is LEVEL COMP holding the level steady. Turn it off for the raw level change.
VC knobs do nothing	Check the stage's VC source on the STAGES page. JACK needs a cable (rack only). SIDECHAIN needs a signal on the plugin's sidechain input.
VCA SOURCE CV does nothing	In the plugin it needs a sidechain signal. In the rack it needs a cable into VCA CV. Also check that VCA DEPTH is above 0.
Track is late against others at 2x	Turn on plugin delay compensation in your DAW. ORIGAMI reports its 46 samples.
Harsh high end	Try QUALITY 2x, lower EMPH, or bring MIX down.
Plugin not listed	Check the VST3 folder (section 7) and rescan in your DAW.
Lopsided waveform or offset	The 8 Hz DC blocker removes offset after the folder. Very low frequencies under heavy SYM still change shape. That is the fold itself.

10. Legal

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Appendix: Back panel patching (ORIGAMI in the JIDAI RACK)

In the JIDAI RACK, ORIGAMI is a 3 U device when open and a 1 U strip when closed. Press **Tab** (or click **BACK** in the rack header) to flip the whole rack around. On the back, ORIGAMI is a 1 U rear plate that carries all 14 of its jacks.



Rear jacks

Group	Jack	Direction	Signal	Notes
INPUT	IN L	in	audio	
INPUT	IN R	in	audio	Follows IN L when only IN L is patched
INPUT	VCA CV	in	CV	0-5 V, used when VCA SOURCE is CV
VC (AUDIO RATE OK)	VC 1, VC 2, VC 3	in	CV	A cable here wins over the stage's VC source
OUTPUT	OUT L, OUT R	out	audio	
HOST (NORMALS)	IN L, IN R	in	audio	Back only. Feeds IN L/R while those are unpatched.
HOST (NORMALS)	OUT L, OUT R	out	audio	Back only. The same signal as OUT L/R.
SIDECHAIN	SC L, SC R	in	audio	Back only. The sidechain input, averaged to mono when both are patched.

Colours and labels

Each jack's ring shows its signal type, and cables take the colour of the jack they come from:

Colour	Signal
Red	Audio
Yellow	CV
Green	Gate / clock
Blue	1 V/oct pitch
Light blue	Linear Hz/V pitch
Purple	S-trigger

Output jacks have a light label. Inputs have a plain label.

Normalled connections

- **IN R ← IN L**: with only IN L patched, both channels get the IN L signal.

- **HOST IN ← into IN L/R:** while IN L/R are unpatched, the back-only HOST IN jacks feed them. Patch the rack's host audio (RACK I/O > HOST IN) into ORIGAMI's HOST IN to process your track while keeping IN L/R free for other sources.
- **OUT ← HOST OUT:** HOST OUT L/R always carry the same signal as OUT L/R, so you can send the output to two places from tidy positions on the plate.
- When you add an ORIGAMI to a rack, the rack patches it for you on the back-only jacks: RACK I/O > HOST IN L/R into ORIGAMI's HOST IN L/R, and ORIGAMI's HOST OUT L/R into RACK I/O > MAIN OUT L/R. IN L/R and OUT L/R stay free. Hold **Shift** while adding to place it without cables.

Cable views

Press **K** to cycle the cable views. The rack header shows which one is lit: **ALL** (every cable as a rope), **HIDE PASS-THRU** (long cables become short stubs tagged with their far end), **SELECTED** (the selected device's cables at full strength) and **HIDE** (plugs only). The rack manual covers patching gestures in full.

Worked example: Stepped Fold (starter rack)

Open the rack's **RACKS** menu and choose **EDM > Stepped Fold**, then press Tab.



The rack has the cables shown above:

1. **RONIN > HOST OUT L / HOST OUT R → ORIGAMI IN L / IN R** (red): RONIN's output, a saw drone gated by BUSHIDO, into the folder.
2. **BUSHIDO > CV C → ORIGAMI VC 1** (yellow): row C of the sequencer sets stage 1's fold for every step.
3. **BUSHIDO > step 5 TRIG → ORIGAMI VC 3** (green): once per loop a trigger kicks stage 3.
4. **ORIGAMI > OUT L / OUT R → RACK I/O > MAIN OUT L / R**: to your DAW.

Try moving the VC 1 cable to VC 2, or patching **RONIN > MG TRI** into **VCA CV** with VCA SOURCE on CV for a slow, breathing drive.

Worked example: Filter Fold FX (starter rack)

FX > Filter Fold FX runs your track through RONIN's resonant filter, swept by its MG, and then through ORIGAMI: RACK I/O > HOST IN L/R → RONIN > HOST IN L/R (back only, feeds RONIN's EXT IN) → EXT IN MONO → VCF → VCA 1 → OUTPUT WET → RONIN > HOST OUT L/R → ORIGAMI > IN L/R → ORIGAMI > OUT L/R → RACK I/O > MAIN OUT L/R. Insert JIDAI RACK on an audio track to hear it.